

Techno-Economic Analysis of Hybrid Hydrogen Fuel-Cell/PV/Wind turbine/Battery/Diesel Energy System for Rural Coastal Community in Western Australia

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Abstract—A Hybrid Renewable Energy System (HRES) is the most feasible and sustainable method for replacing the current conventional diesel generation system with the aim of decarbonization, as well as providing a continuous and stable power supply to residents in rural areas worldwide. Hydrogen energy storage technology is a promising method for producing clean electricity and reducing CO₂ emissions compared to solar PV and wind turbine power systems. This study presents the techno-economic analysis of the Fuel cell / PV / Wind / Battery / Diesel HRES in a rural region in Western Australia using HOMER. Three system configurations with different penetration levels of hydrogen generation have been compared in terms of financial costs and decarbonization. The results show that HRES integrated with hydrogen technology has a good performance in reducing CO₂ emissions. The high penetration of the fuel cell HRES results in a reduction of 653 kg/year. However, costs such as Levelized Cost of Electricity (LCOE) and Net Present Cost (NPC) are still not as price-competitive as HRES without a hydrogen fuel cell. The optimal off-grid design identified in this case study is the scenario involving an optimal Fuel cell/PV/Wind turbine/Diesel/Battery HRES with an NPC of \$4.580 million and an LCOE of \$0.5238/kWh for energy costs.

Keywords— renewable energy, hybrid off-grid, hydrogen, CO₂ emissions, carbon management

I. INTRODUCTION

The demand for energy is increasing significantly due to global population growth, industrialization, and economic expansion. Approximately 70% of electricity is generated using conventional, depleting fossil fuels [1,2]. Moreover, the conventional fossil fuel-based power generation system produces a substantial amount of greenhouse gases (GHGs), including carbon dioxide, which is the primary contributor to global warming [3]. Consequently, developing power generation systems that leverage various forms of renewable energy (RE) sources to replace traditional fossil fuels is a significant research direction in terms of safeguarding the global environment and promoting the sustainability of energy systems.

Renewable energy sources, such as solar, wind, hydro, and biomass, currently represent the leading technologies for renewable power generation, with the aim of achieving a more environmentally friendly and sustainable future [4]. The utilization of RE power generation technology can effectively reduce greenhouse gas emissions and reduce the energy industry's reliance on fossil fuels. However, RE power

generation technologies, such as PV and wind turbine systems, face a notable challenge of intermittency, which can lead to power system instability. A popular solution to address this intermittency in RE power systems is the integration of a hybrid renewable energy system (HRES), which combines various types of RE power generation systems and energy storage solutions to ensure the continuity and stability of energy supply.

Unlike the transmission-distribution system in cities, using the grid to supply electricity in remote areas is not the best choice due to high construction costs, maintenance expenses, and operational costs. Consequently, self-sufficient off-grid diesel power systems are commonly used in regions far from urban areas with small populations [5,6]. Furthermore, rural communities serve as suitable locations to showcase HRES instead of conventional diesel power systems, given the environmental concerns, unlike urban areas with grid power systems [7]. Additionally, it is effective to integrate a large-capacity energy storage system to manage load fluctuations in remote area HRES [8].

Hydrogen energy storage technology offers an alternative method by utilizing hydrogen as the medium and fuel cells as the main technology to store excess electricity generated by HRES and provide energy in case of unsatisfied demand for electricity generation. HRES with fuel cell technology benefits the system by reducing the intermittency of PV and wind turbine generation technology, thanks to its features of continuity and stability.

Abdin and Merida simulated the performance of a fuel cell HRES in five selected locations with different weather and load conditions [9]. The levelized cost of energy (LCOE) varied from \$0.52 to \$0.95 per kWh, which indicated that the fuel cell HRES is not cost-effective, resulting in high values for NPCs and LCOEs [9]. Akhtari and Baneshi conducted a study on the fuel cell HRES installed at two locations in Iran [10]. The simulated results show that HRES, integrating hydrogen technology, is feasible in the Bandar Abbas region when considering LCOE [10]. However, NPCs from the simulated results remain high in most cases [10]. Rezk et al. simulated the fuel cell/PV/battery HRES in NEOM city, Saudi Arabia [11]. The performance indicates that the hydrogen technology in HES is viable in this case, with a competitive LCOE of \$0.126 per kWh [11].

HOMER Pro optimization software is utilized in this study for simulating and modeling the renewable energy system, thanks to its flexible accessibility and high efficiency [12]. Consequently, this study pursues four main objectives:

- Optimal design of the Fuel Cell/PV/Wind turbine/Battery/Diesel HRES in a rural coastal community in Western Australia.
- Techno-economic analysis of Fuel Cell/PV/Wind turbine/Battery/Diesel HRES in a rural coastal community in Western Australia.
- Evaluating the feasibility of integrating a hydrogen energy storage system to support the community's energy load in a remote region in Western Australia.
- Calculating the reduction in GHG emissions achieved by the HRES integrating hydrogen energy storage technology.

II. DATA AND METHODOLOGY

A. Solar and wind potential at the selected location in WA

Necessary weather data, such as solar radiation and average wind speed for the selected location (25.9299° S latitude and 113.5381° E longitude), were extracted from the NASA Atmospheric Science Data Center in the HOMER database [13]. The scaled annual average radiation for the selected location is 6.19 kWh/m²/day [13]. Moreover, monthly clearness indices and daily solar radiation are depicted in Fig. 1. The highest solar radiation occurs in June (winter), while the lowest is in December (summer). The scaled annual average wind speed for the selected location is 8.12 m/s. Monthly average wind speeds are presented in Fig. 2. The highest average wind speed is observed in January (summer), while the lowest is in May (winter).

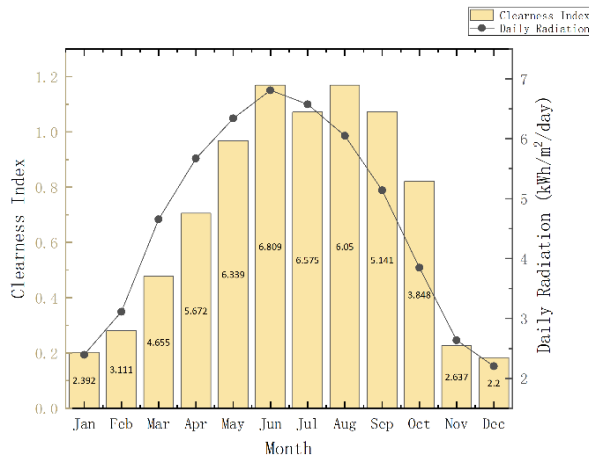


Fig. 1. Monthly radiation and Clearness index of the studied region

It is clear that solar and wind speed RE potentials can be integrated into HRES at the selected location to provide a sustainable way to fulfill local power demand as well as reduce the emission of GHGs with a high penetration of RE.

The RE potentials of solar radiation and average wind speed have different peak seasons, which ensures the abundant RE resources in summer and winter times of the selected region, and HRES providing stable and continuous power to the local residents.

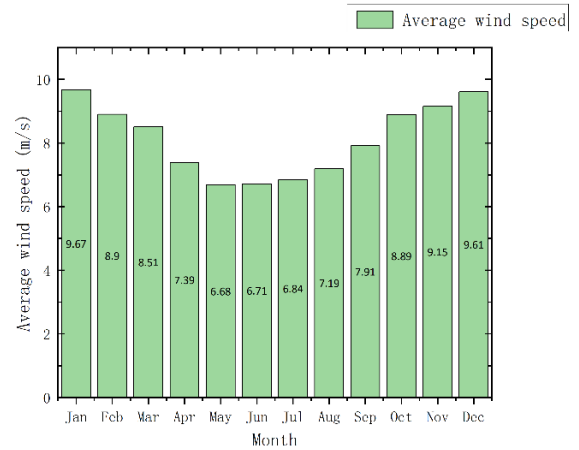


Fig. 2. Monthly average wind speed

B. Load profile

The mainstay of the studied area is tourism. The impact of tourism on the local load profile can result in increased electricity consumption from late afternoon to late night in the commercial mode. Consequently, the load profile of this community is based on a combination of the commercial and community load profiles, integrating the impact of tourism. The daily power load for the studied community is set at 1441 kWh.

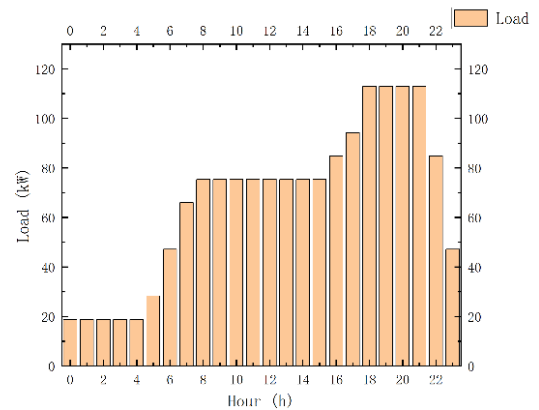


Fig. 3. Load profile

Fig. 3 shows the daily load curve of the community in the selected area. The peak electricity consumption period is in the evening from 18 to 21, primarily due to highly intensive commercial and tourism activities, while the time period of low power demand in the studied area is from 0:00 a.m. to 5:00 a.m.

C. Ecomic considerations for HRES design

Net present cost (NPC)

Net present cost (NPC) is defined as the net sum of the combined initial installation costs used over the project life and all operating and maintenance (O&M) costs required over the project life, minus the total value of revenue earned in each year of the project. It is also a popular tool for analyzing economic feasibility. The equation below is programmed in HOMER Pro to calculate the NPC [14]:

$$NPC = \sum_{n=1}^t R_i (C_{Sal} + C_{Cap} + C_{Rep} + C_{O\&M} + C_{Fuel}) \quad (1)$$

where R_i is each year discount rate; n is the years in period of the designed project; C_{Sal} is the savage cost; C_{Cap} is the capital cost of the project in year 0; C_{Rep} is the replacement cost; $C_{O\&M}$ is the operation and maintenance cost; C_{Fuel} is the fuel cost.

- *Capital recovery factor function*

Capital recovery factor function is considered as the specific operating year's ratio for calculating the cost at that specific year [14]. The calculated equation is shown below [15]:

$$CRF(i, N) = \frac{i(1+i)^N}{(1+i)^N - 1} \quad (2)$$

where, i is the annual real discount rate; N is the exact operating number of the year.

- *Annualized cost*

The annualized cost is defined as the annualized net present value cost of the cash flow series associated with costs that will be spent in each year of the specified project simulation time [14]. The annualized cost is calculated by the equation shown below [14]:

$$C_{ann} = CRF(i, N) NPC \quad (3)$$

- *Levelized cost of energy (LCOE)*

The Levelized Cost of Energy (LCOE) is a technical metric that quantifies the average expenditure required to generate each kilowatt-hour (kWh) of electricity within the simulation system, measured in (\$/kWh) [14]. The choice of currency can be adapted to the local currency of the simulation location, but it is common practice to employ the US dollar as the standard unit for all computations across various reports. The calculation of LCOE is precisely defined as the difference between the total annualized cost and the total annualized thermal cost, which is allocated to fulfill the thermal load within the system, divided by the total supplied electrical load. The formula for computing LCOE is depicted below [14]:

$$LCOE = \frac{C_{ann} - c_b Load_T}{Load_E} \quad (4)$$

where C_{ann} is the annualized cost; c_b is boiler marginal cost parameter; $Load_T$ is the thermal load power cost per year; $Load_E$ is the electrical load power cost per year. If the HES has the diesel generator, the HES will need to calculate the thermal load cost every year. Furthermore, the LCOE will need to be minus that part of cost as well.

D. Modelling and sizing of HRES

- *Solar PV system*

The power generation of the PV system is determined by the solar irradiance and the module characteristics. The power production of simulated PV module model in this design can be expressed as [16]:

$$P_{PV} = \eta_{PV} R_t A_{PV} \quad (5)$$

where η_{PV} is the overall PV efficiency; n is the years in period of the designed project; R_t is the solar radiation (W/m^2); A_{PV} is the PV generation area.

PARAMETERS OF SOLAR PV SYSTEM

Parameter	Value
Capacity (kW)	700
Capital (\$)	1,575,000
Replacement (\$)	1,575,000
O&M (\$)	625
Derating factor (%)	80
Lifetime (year)	20

The specific parameters of the selected PV system for the optimal HRES are presented in Table I. The currency used in this project is the US dollar (\$). Additionally, the lifetime of the PV module system is 25 years. The system control strategy chosen is the cycle charging dispatch strategy. A simplified diagram of this HRES is depicted in Fig. 4.

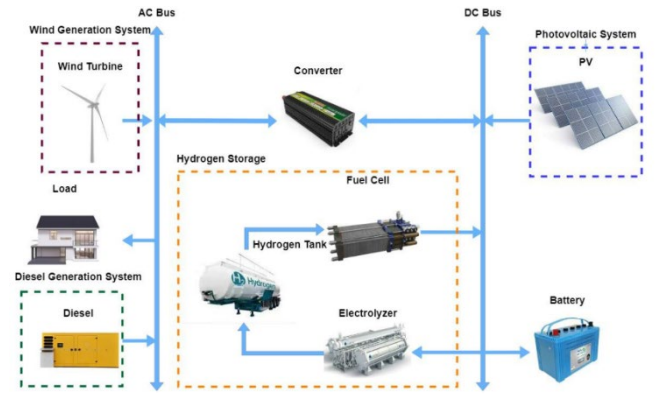


Fig. 4. General schematics of HRES

- *Wind turbine system*

The wind speed at the hub height has a direct relationship with the actual output power. Consequently, using the equation below, the recorded wind speed at the reference height should be converted to the appropriate hub height [17]:

$$V_h(t) = V_{ref}(t) \frac{\ln(H/Z_0)}{\ln(H_{ref}/Z_0)} \quad (6)$$

where V_h is velocity at hub height; V_{ref} is velocity at reference height; Z_0 is the length of surface roughness; H is hub height of wind turbine; H_{ref} is the reference height of wind turbine.

The selected optimal wind turbine module system and its specific parameters are shown in Table II.

PARAMETERS OF WIND TURBINE SYSTEM

Parameter	Value
Capacity (kW)	230
Capital (\$)	95,000
Replacement (\$)	81,000
O&M (\$)	202,500

PARAMETERS OF HYDROGEN ENERGY STORAGE SYSTEM

Components	Parameter	Value
Fuel cell	Capacity (kW)	100
	Capital (\$)	3,000,000
	Replacement (\$)	3,000,000
	O&M (\$)	17,520
Electrolyser	Capacity (kW)	348
	Capital (\$)	696,000
	Replacement (\$)	696,000
	O&M (\$)	450
Hydrogen tank	Capacity (kg)	312
	Capital (\$)	1000
	Replacement (\$)	800
	O&M (\$)	250

- *Hydrogen energy storage systems*

Hydrogen energy storage system consists of fuel cell, electrolyser, hydrogen tank. Table III demonstrates the essential parameters of hydrogen energy storage system in this optimal HRES.

- *Diesel generator*

Table IV presents the key parameters of diesel generator in the optimal HRES.

PARAMETERS OF DIESEL GENERATOR

Parameter	Value
Capacity (kW)	365
Capital (\$)	56,000
Replacement (\$)	40,000
O&M (\$)	61,320

- *Converter*

The converter in this optimal HRES model is used to convert the DC and AC currents, as the different power generation devices or systems producing AC and DC two types of currents. Table V shows the fundamental parameters of the converter in this HRES.

PARAMETERS OF CONVERTER

Parameter	Value
Capacity (kW)	98.4
Capital (\$)	29,520
Replacement (\$)	29,520

PARAMETERS OF BATTERY

Parameter	Value
Capacity (strings)	224
Capital (\$)	71,680
Replacement (\$)	71,680
O&M (\$)	200
Throughput (kWh)	800

- *Battery*

The available electricity of the system as a function of time and the load demand of the system as a function of time are the two essential parameters for determining the state of charge (SOC) or discharging of the battery system. The SOC as a function of time can be calculated using the equation below [18]:

$$SOC(t) = \begin{cases} SOC(t-1) + \frac{\eta_b \Delta_t (P_s - P_d)}{C_b}, & \text{charging} \\ SOC(t-1) + \frac{\eta_b \Delta_t (P_d - P_s)}{C_b}, & \text{discharging} \end{cases}$$

where η_b is battery roundtrip efficiency; C_b is battery nominal capacity; P_s is total available power; P_d is demand load power; Δ_t is the time interval. The parameters of the battery in this optimal HRES are shown in Table VI.

III. RESULTS

This study is focused on the techno-economic analysis of a Fuel cell / PV / Wind turbine / Battery / Diesel Hybrid Renewable Energy System (HRES) in a remote area of Western Australia, known for its abundant wind and solar radiation renewable energy resources.

Three different system configurations have been analyzed for their electrical and economic performances: PV/Wind/Diesel/Battery HRES, the optimal Fuel cell/PV/Wind/Diesel/Battery HRES, and the high penetration Fuel cell/PV/Wind/Diesel/Battery HRES. The optimal HRES configuration is selected based on the minimum cost of the integrated HRES from the algorithm in HOMER. The high penetration Fuel cell HRES is the configuration that increases the percentage of power generated by the fuel cell in hydrogen technology by ensuring the fuel cell operates for 8,760 hours throughout the year in HOMER. Among all the simulated HRES configurations in this case study, the PV/Wind/Diesel/Battery HRES has the lowest LCOE.

A. Electrical comparison

The total generated electricity of each part percentage in different system configurations are demonstrated in Fig. 5.

The highest energy generation comes from the high penetration fuel cell HRES, producing 2,112,459 kWh/year, with 18.2% from the fuel cell, totalling 384,493 kWh/year. The optimized Fuel cell/Wind/PV/DG/Battery HRES produces a total of 1,828,194 kWh/year, while the lowest power generation is provided by the PV/Wind/DG/Battery HRES, with 1,742,674 kWh/year. The solar PV system plays the primary role in generating power for all three HRES configurations. Additionally, the wind turbine system produces the second-highest amount of electricity in the simulated results in HOMER Pro. Concerning the non-renewable fraction, diesel contributes only a small percentage of power for the selected region. Meanwhile, the renewable fraction (RF) of all three HRES configurations is high, with the highest RF being 99.8% for the high penetration Fuel cell HRES, and the lowest being 97.0% for the PV/Wind/Battery/DG HRES.

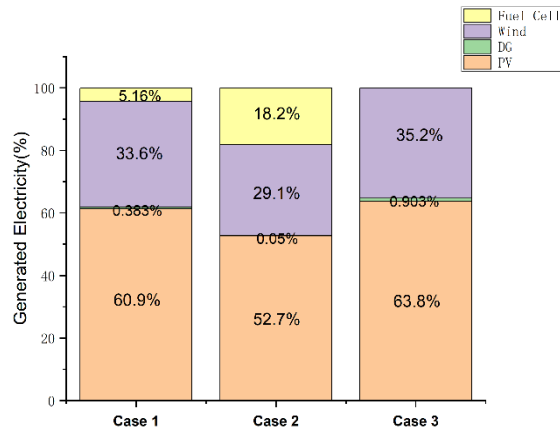


Fig. 5. Power generation and the renewable fractions of three configurations: Case 1 optimal Fuel cell/PV/Wind/Diesel/Battery HRES; Case 2 high penetration Fuel cell/PV/Wind/Diesel/Battery HRES; Case 3 PV/Wind/Diesel/Battery HRES

B. Economic comparison

LCOE is the main parameter in the economic comparison of the system configurations, reflecting the highest LCOE of \$0.7396/kWh in the high penetration Fuel cell HRES and the lowest LCOE of \$0.3935/kWh in the PV/Wind/Diesel/Battery HRES. Detailed economic comparisons are provided in TABLE VII.

ECONOMIC COMPARISON OF THREE SYSTEM CONFIGURATIONS

System Architecture	NPC (million \$)	LCOE (\$/kWh)	Operating cost (million \$)
Optimal Fuel cell HRES	4.580	0.5238	0.090
High penetration Fuel cell HRES	6.462	0.7396	0.207
PV/Wind/DG/Battery HRES	3.407	0.3935	0.086

C. GHG emissions comparison

Decarbonization is the critical objective for replacing the conventional diesel-based power system with HRES. A comparison of GHG emissions among different system architectures is presented in Table IX and Fig. 6.

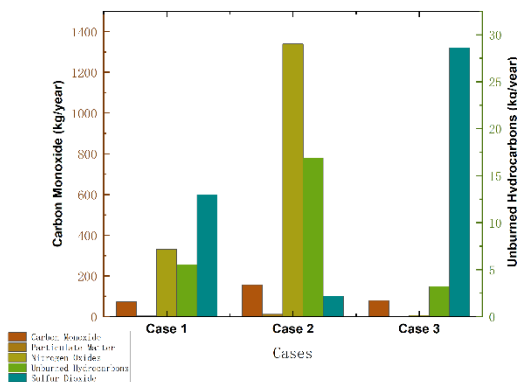


Fig. 6. GHG emissions generated by three configurations: Case 1 optimal Fuel cell/PV/Wind/Diesel/Battery HRES; Case 2 high penetration Fuel cell/PV/Wind/Diesel/Battery HRES; Case 3 PV/Wind/Diesel/Battery HRES

The GHG emissions from three system configurations demonstrate the lowest CO₂ emission of 653 kg/year with the high penetration Fuel cell HRES, while the highest CO₂ emission of 11,660 kg/year is observed with the PV/Wind/Diesel/Battery HRES. However, the high penetration Fuel cell HRES also results in the highest NO_x emissions of 1,339 kg/year.

GHG EMISSIONS COMPARISONS OF THREE SYSTEM CONFIGURATIONS

GHG type	Optimal Fuel cell HRES	High penetration Fuel cell HRES	PV/Wind/DG/Battery HRES
CO ₂ (kg/year)	5,232	653	11,660
CO _x (kg/year)	72.7	156	79.3
Unburned Hydrocarbons (kg/year)	5.53	16.9	3.221
Particulate Matter (kg/year)	2.92	11.3	0.317
SO _x (kg/year)	13.0	2.18	28.6
NO _x (kg/year)	331	1,339	6.35

IV. DISCUSSION

A. Optimal system configuration

The optimal off-grid design identified in this case study is the scenario of the optimal Fuel cell/PV/Wind turbine/Diesel/Battery HRES with an NPC of \$4.580 million and an LCOE of \$0.5238/kWh for energy cost. The parameters of components in the designed HRES are presented in TABLE I to TABLE IV. The main power generated by the solar PV is 700 kW. Moreover, the renewable fraction of the optimal system is 98.7%.

The comparative results for electricity, economics, and GHG emissions demonstrate that the Fuel cell/PV/Wind turbine/Diesel/Battery HRES is the optimal system configuration, considering the integration of hydrogen energy storage technology and the lowest LCOE. However, the PV/Wind/Diesel/Battery HRES has the lowest LCOE, NPC, and operating cost among the three different configurations. The optimal HRES strikes a balance between financial considerations and the reduction of GHG emissions. It is the optimal HRES with a fuel cell but not the most optimized HRES with the lowest LCOE, as the conventional diesel power system and PV/Wind/Diesel/Battery HRES have lower LCOEs than the optimal HRES. However, the conventional diesel power system and PV/Wind/Diesel/Battery HRES emit a greater amount of GHGs than HRES integrated with a fuel cell, aiming for decarbonization.

B. Hydrogen energy storage technology

The lowest LCOE of 0.3935 \$/kWh is associated with the HRES without hydrogen energy storage technology, while the highest CO₂ emission of 11,660 kg/year is observed in the HRES without hydrogen generation. The optimal fuel cell HRES has an LCOE of 0.5238 \$/kWh and a CO₂ emission of 5,232 kg/year. For higher penetration of hydrogen fuel cell (18.2%), it has the highest LCOE of 0.7396 \$/kWh and the highest NPC of 6.462 million dollars, which is not price-competitive. However, the CO₂ emission is 653 kg/year, perfectly aligning with the aim of power system decarbonization.

In the economic comparison sector, the high costs of LCOE, NPC, and operating costs are the primary constraints to integrating hydrogen fuel cell technology into HRES, despite the excellent performance in decarbonization as indicated by the GHG emission comparison. Furthermore, the increased energy production by the fuel cell results in higher costs for HRES. With the advancement of fuel cell technology and the reduction in the cost of each component, the application of hydrogen energy storage technology in HRES holds great promise for the future.

It is further concluded from the results that integrating the hydrogen energy storage system with other renewable energy resources such as solar and wind helps reduce costs. Additionally, the results demonstrate that the hydrogen generation system contributes to the reduction of carbon dioxide emissions..

V. CONCLUSION

The off-grid HRES is a sustainable and feasible option for remote areas to supply local power demand while achieving the goal of power system decarbonization. This study discusses three HRES models with different penetrations of hydrogen generation technology in a rural community in WA. The optimal HRES with a hydrogen fuel cell has an LCOE of 0.5238 \$/kWh, NPC of 4.580 million dollars, and CO₂ emissions of 5,232 kg/year.

The high penetration of fuel cell HRES presents the lowest CO₂ emissions of 653 kg/year but also the highest LCOE of 0.7396 \$/kWh and the highest NPC of 6.462 million dollars. Moreover, the PV/Wind/Diesel/Battery HRES has the lowest LCOE of 0.3935 \$/kWh and the lowest NPC of 3.407 million dollars, while exhibiting the highest CO₂ emissions of 11,660 kg/year. Consequently, the hydrogen generation system has an outstanding performance in terms of decarbonization. However, the constraint of hydrogen fuel cell is cost. With a higher penetration of hydrogen fuel cell integrated into HRES, more CO₂ emissions can be avoided. At the same time, the cost increases with a high penetration of hydrogen fuel cell. From the simulated results, the HRES integrated with hydrogen technology is still not cost-competitive.

Future work's primary focus is on implementing a standalone HRES configuration that balances the economic aspects and CO₂ emissions with novel optimization algorithms in different simulation software to explore the potential of hydrogen generation systems and the feasibility of integrating hydrogen technology into off-grid HRES.

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